15. Because $N[P]$ is a normal subgroup of $N[N[P]]$, conjugation of P by an element of $N[N[P]]$ yields a subgroup of $N[P]$ that is a Sylow p-subgroup of G, and also of $N[P]$. Such a Sylow p-subgroup must be conjugate to P under conjugation by an element of $N[P]$ by Theorem 36.10, and must therefore be P because P is a normal subgroup of $N[P]$. Thus P is invariant under conjugation by every element of $N[N[P]]$, so $N[N[P]]$ is contained in $N[P]$. Because $N[P]$ is contained in $N[N[P]]$ by definition, we see that $N[N[P]] = N[P]$.

16. By Theorem 36.8, H is contained in a Sylow p-subgroup K of G. By Theorem 36.10, there exists $g \in G$ such that $gKg^{-1} = P$. Consequently $gHg^{-1} \leq P$.

17. The divisors of $(35)^3$ that are not divisible by 5 are 1, 7, 49, and 343, which are congruent to 1, 2, 4, and 3 respectively modulo 5. By Theorem 36.11, there is only one Sylow 5-subgroup of a group of order $(35)^3$, and it must be a normal subgroup by the argument in Exercise 12.

18. The divisors of 255 that are not multiples of 17 are 1, 3, 5, and 15. Of these, only 1 is congruent to 1 modulo 17. By Theorem 36.11, there is only one Sylow 17-subgroup of a group of order 255, and it must be a normal subgroup by the argument in Exercise 12. Thus no group of order 255 can be simple.

19. The divisors of $p^r m$ that are not divisible by p are 1 and m. Because $m < p$, of these two divisors of $p^r m$, only 1 is congruent to 1 modulo p. By Theorem 36.11, there is a unique Sylow p-subgroup of a group of order $p^r m$ where $m < p$, and this must be a normal subgroup by the argument in Exercise 12. Thus such a group cannot be simple.

20. **a.** As a G-set under conjugation, we have $G_G = \{g \in G \mid gxg^{-1} = x \text{ for all } x \in G\} = \{g \in G \mid gx = xg \text{ for all } x \in G\} = Z(G)$ by definition of $Z(G)$.

b. Let G be a nontrivial p-group, so that $|G| = p^r$ for $r \geq 1$. By Theorem 36.1, we see that $|G| \equiv |G_G| \pmod{p}$. Thus p is a divisor of $|G_G|$, and hence is a divisor of $|Z(G)|$ by Part(**a**). Thus $Z(G)$ is nontrivial.

21. We proceed by induction on n. If $n = 1$, the statement is obviously true, for $H_0 = \{e\}$ and the entire group $G = H_1$ are the required subgroups. If $n = 2$, the subgroups are supplied by Theorem 36.8. Suppose the statement is true for $n = k$, and let G have order p^{k+1}. Let $Z(G)$ have order p^j where $j \geq 1$ by Exercise 20. If $j = k+1$, then G is abelian and the subgroups provided by Theorem 36.8 are all normal subgroups of G and we are done. If $j < k+1$, apply the induction hypothesis to $Z(G)$ to find its desired normal subgroups $H_0 < H_1 < \cdots < H_{j-1} < H_j = Z(G)$. Then form the factor group $G/Z(G)$ which has order p^{k+1-j} and find normal subgroups $K_1 < K_2 < \cdots < K_{k+1-j}$ of $G/Z(G)$ where the order of K_i is p^i. If $\gamma : G \to G/Z(G)$ is the canonical homomorphism, then $H_{j+i} = \gamma^{-1}[K_i]$ is a normal subgroup of G of order p^{i+j}, and all these subgroups H_i form the desired chain of normal subgroups of G.

22. Let P be a normal p-subgroup of G. By Theorem 36.8, there exists a Sylow p-subgroup H of G containing P. By Theorem 36.20, every Sylow p-subgroup of G is of the form gHg^{-1} for some $g \in G$. Because $gPg^{-1} = P$, we see that P is contained in every Sylow p-subgroup of G.

37. Applications of the Sylow Theory

1. **a.** The conjugate classes are $\{\rho_0\}, \{\rho_2\}, \{\rho_1, \rho_2\}, \{\mu_1, \mu_2\}$, and $\{\delta_1, \delta_2\}$.

b. The class equation is $8 = 2 + 2 + 2 + 2$.

2. As p-groups, groups of these orders are not simple by Theorem 36.8: 4, 8, 9, 16, 25, 27, 32, 49.

As groups of order pq, groups of these orders are not simple by Theorem 37.7: 6, 10, 14, 15, 21, 22, 26, 33, 34, 35, 38, 39, 46, 51, 55, 57, 58.

As groups of order $p^r m$ with $m < p$, groups of these orders are not simple by Exercise 19 of Section 36: 18, 20, 21, 28, 42, 44, 50, 52, 54.

The text showed that groups of these orders are not simple: 30, 36, 48.

Order 12: Such a group has either 1 or 3 Sylow 2-subgroups of order 4 and either 1 or 4 Sylow 3-subgroups of order 3. To have 3 subgroups of order 4 and 4 subgroups of order 3 would require at least 4 elements of order divisible by 2 and at least 8 elements of order 3, which would require 12 elements other than the identity. Thus there is either only one subgroup of order 4 or only one of order 3, which must be normal.

Order 24: Such a group has either 1 or 3 subgroups of order 8 and either 1 or 4 subgroups of order 3. If there is a unique subgroup of either order, we are done. Suppose that H and K are different subgroups of order 8. By Lemma 37.8, $H \cap K$ must have order 4, and is normal in both H and K, being of index 2. Thus $N[H \cap K]$ contains both H and K so it has order a multiple > 1 of 8 and a divisor of 24. Hence $N[H \cap K]$ is of order 24 and $H \cap K$ is a normal subgroup.

Order 40: Theorem 36.11 shows that there is a unique subgroup of order 5, which must be normal.

Order 45: Theorem 35.11 shows that there is a unique subgroup of order 9, which must be normal.

Order 56: Such a group has either 1 or 7 subgroups of order 8 and either 1 or 8 subgroups of order 7. If there is a unique subgroup of either order, we are done. Eight subgroups of order 7 require 48 elements of order 7, and 7 subgroups of order 8 require at least 8 elements of order divisible by 2, which is impossible in a group of 56 elements.

All orders from 2 to 59 that are not prime have been considered. We know that A_5, which has order 60, is simple.

3. T T F T T T T T F F

4. By Theorem 36.11, a group G of order $5 \cdot 7 \cdot 47$ contains a unique subgroup H of order 47, which must be normal in G. By the same arguments, there exist unique normal subgroups K and L of orders 7 and 5 respectively. By Lemma 37.8, LK has order 35 because $L \cap K = \{e\}$. By the proof of Lemma 37.8, $(LK)H$ has order $35 \cdot 47$, so $(LK)H = G$. Now LK must be the unique subgroup of G of order 35, because another subgroup would lead to subgroups of orders 7 and 5 other than L and K, which is impossible. By Lemma 37.5, G is isomorphic to $LK \times H$ and consequently to $L \times K \times H$ which is abelian and cyclic.

5. A group of order 96 has either 1 or 3 subgroups of order 32. If there is only one such subgroup, it is normal and we are done. If not, let H and K be distinct subgroups of order 32. By Lemma 37.8, $H \cap K$ must have order 16, and is normal in both H and K, being of index 2. Thus $N[H \cap K]$ has order a multiple > 1 of 32 and a divisor of 96, so the order must be 96. Thus $H \cap K$ is normal in the whole group.

6. A group G of order 160 has either 1 or 5 subgroups of order 32 and either 1 or 16 subgroups of order 5. If there is only one of order 32 or only one of order 5, it is a normal sbgroup and we are done. Let us suppose that this is not the case. Let H and K be distinct subgroups of order 32. By Lemma

37.8, $H \cap K$ must have order either 16 or 8. If $|H \cap K| = 16$, then it is normal in both H and K, so $N[H \cap K]$ has order a multiple > 1 of 32 and a divisor of 160, so $N[H \cap K] = G$ and $H \cap K$ is a normal subgroup of the group G. If $|H \cap K| = 8$, then HK has order $(32)(32)/8 = 128$ by Lemma 37.8, so G has at least 127 elements of order divisible by 2. Then 16 subgroups of order 5 would contribute 64 elements of order 5, and $127 + 64 > 160$, which is impossible. Thus G is not simple.

7. By Example 37.12, a group G of order 30 has a normal subgroup of order 5 or of order 3. Suppose that G has a normal subgroup H of order 5. Then G/H is a group of order 6, which has a normal subgroup K of order 3 by Sylow theory. If $\lambda : G \to G/H$ is the canonical homomorphism, then $\lambda^{-1}[K]$ is a normal subgroup of G of order $3 \cdot 5 = 15$. If G has no normal subgroup of order 5, then it has a normal subgroup N of order 3, so G/N has order 10 and has a normal subgroup L of order 5. Applying to L the inverse of the canonical homomorphism mapping G onto G/N gives a normal subgroup of G of order 15.

8. a. We have $\tau\sigma\tau^{-1}(\tau(a_i)) = \tau\sigma(a_i) = \begin{cases} \tau(a_{i+1}) & \text{if } i < m, \\ \tau(a_1) & \text{if } i = m. \end{cases}$

For any element b not of the form $\tau(a_i)$, we have $\tau\sigma\tau^{-1}(b) = \tau\tau^{-1}(b) = b$. Thus $\tau\sigma\tau^{-1}$ has the desired action on each element of $\{1, 2, \cdots, n\}$.

b. Let $\sigma = (a_1, a_1, \cdots, a_m)$ and $\mu = (b_1, b_2, \cdots, b_m)$ be cycles of length m in S_n. Let τ be any permutation in S_n such that $\tau(a_i) = b_i$. Part**a** then shows that $\tau\sigma\tau^{-1} = \mu$.

c.Let σ and μ be products of s disjoint cycles with the ith cycle in each product of length r_i. Let τ be any permutation in S_n that carries the jth element of the ith cycle of σ into the jth element of the ith cycle of μ for $1 \le j \le r_i$ for each i where $1 \le i \le s$. (That is, if we write μ directly under σ and erase all the parentheses at the ends of the cycles, we get a 2-rowed notation for the action of τ on the elements moved by σ.) Fill in the action of τ on the other elements in any way that yields an element of S_n. Repetition of the computation in Part(**a**) shows that $\tau\sigma\tau^{-1} = \mu$.

d. Let σ be any permutation in S_n. Express σ as a product of disjoint cycles, supplying cycles of length 1 for all elements not moved by σ. The sum of the lengths of these cycles is then n, and the sum yields a partition of n. By Part(**c**), σ is conjugate to any other permutation that can be expressed similarly as a product of disjoint cycles that yield the same partition of n. The preceding parts of this exercise show that this correspondence between partitions of n and conjugate classes is one to one.

e. $1 = 1$, so $p(1) = 1$

$2 = 2 = 1 + 1$, so $p(2) = 2$

$3 = 3 = 1 + 2 = 1 + 1 + 1$, so $p(3) = 3$

$4 = 4 = 1 + 3 = 2 + 2 = 1 + 1 + 2 = 1 + 1 + 1 + 1$, so $p(4) = 5$

If $p_1(n)$ is the number of partitions of n having 1 as a summand, then $p_1(n) = p(n-1)$ for $n \ge 2$, for all such partitions of n can be obtained by putting $+1$ after each partition of $n - 1$. Thus for $n \ge 2, p(n) =$(number of partitions without 1 as a summand) $+ p(n-1)$.

$5 = 5 = 2 + 3$, so $p(5) = 2 + p(4) = 2 + 5 = 7$

$6 = 6 = 2 + 4 = 3 + 3 = 2 + 2 + 2$, so $p(6) = 4 + p(5) = 4 + 7 = 11$

$7 = 7 = 2 + 5 = 3 + 4 = 2 + 2 + 3$, so $p(7) = 4 + p(6) = 4 + 11 = 15$

9. Using Exercise 8, we see the conjugate classes are
$\{\iota\}$,
$\{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$,

$\{(1,2)(3,4),\ (1,3)(2,4),\ (1,4)(2,3)\}$,
$\{(1,2,3),(1,2,4),(1,3,4),(1,3,2),(1,4,2),(1,4,3),(2,3,4),(2,4,3)\}$,
$\{(1,2,3,4),(1,2,4,3),(1,3,2,4),(1,3,4,2),(1,4,2,3),(1,4,3,2)\}$
The class equation is $24 = 1 + 6 + 3 + 8 + 6$.

10. *Class Equation for S_5* : Rather than list as we did for Exercise 9, we use combinatorics for S_5. All we want is the class equation.

There are $\binom{5}{2} = 10$ transpositions.

There are $\binom{5}{2}\binom{3}{2}/2 = (10)(3)/2 = 15$ products of two disjoint transpositions.

There are $\binom{5}{3} \cdot 2 = 10 \cdot 2 = 20$ cycles of length 3.

By the previous computation, there are $20 \cdot 1 = 20$ products of disjoint cycles of lengths 3 and 2.

There are $5 \cdot 6 = 30$ cycles of length 4. (Five choices for the number not moved by the cycle, and then 6 cycles using the remaining four numbers as shown in Exercise 9.)

There are $4 \cdot 3 \cdot 2 \cdot 1 = 24$ cycles of length 5. (Put 1 in left position, and fill the remaining four positions in 4! ways.)

The class equation is $120 = 1 + 10 + 15 + 20 + 20 + 30 + 24$.

Class Equation for S_6 : We continue to use combinatorics.

There are $\binom{6}{2} = 15$ transpositions.

There are $\binom{6}{2}\binom{4}{2}/2 = (15)(6)/2 = 45$ products of two disjoint 2-cycles.

There are $\binom{6}{2}\binom{4}{2}\binom{2}{2}/6 = (15)(6)(1)/6 = 15$ products of three disjoint 2 cycles.

There are $\binom{6}{3} \cdot 2 = 20 \cdot 2 = 40$ cycles of length 3

There are $40 \cdot \binom{3}{2} = 120$ products of a 3-cycle and a disjoint 2-cycle. (40 choices for the 3 cycle by the last computation, times $\binom{3}{2}$ choices for a tranposition from the remaining three elements.)

There are $(40 \cdot 2)/2 = 40$ products of two disjoint 3-cycles. (Choose one of the 40 3-cycles, there are only two choices for the other 3 cycle, and divide by 2 because the order of their choice doesn't matter.)

There are $\binom{6}{4} \cdot 6 = 15 \cdot 6 = 90$ 4-cycles. (Choose 4 of the 6 numbers as those to be moved, and there are 6 different 4-cycles moving on them as shown in the solution of Exercise 9.)

There are $90 \cdot 1 = 90$ products of a 4-cycle and a disjoint 2-cycle. (Choose 1 of the 90 4-cycles, and there is only one choice for the disjoint 2-cycle.)

There are $\binom{6}{5} \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 6 \cdot 24 = 144$ 5-cycles. (Choose the 5 elements to be moved, list the smallest of them at the left, and then fill in the remaining 4 position in 4! ways.)

There are $5! = 120$ 6-cycles. (Put 1 in left position, and fill the remaining 5 positions in 5! ways.)

The class equation is $720 = 1 + 15 + 45 + 15 + 40 + 120 + 40 + 90 + 90 + 144 + 120$.

11. Exercise 8 shows that the number of conjugate classes in S_n is the number $p(n)$ of partitions of n. By Theorem 11.12, the number of abelian groups of order p^n is also the number of partitions of n; it is the number of ways that p^n can be split up into a product of powers of p, where the order of the product doesn't matter. It is determined by how split the exponent n into a sum of exponents for the factors; that is, by how to partition n.